

Quiz (Habanero)

Q1. Simplify $\sqrt{48} + \sqrt{18}$

- (A) $4\sqrt{3} + 3\sqrt{2}$
- (B) $6\sqrt{2} + 4\sqrt{3}$
- (C) $4\sqrt{3} + 3\sqrt{3}$
- (D) $2\sqrt{6} + 3\sqrt{2}$
- (E) $6\sqrt{3} + 2\sqrt{2}$

Q2. The temperature in a greenhouse is $\sqrt{80}$ degrees higher than outside. Simplify this expression

- (A) $4\sqrt{5}$
- (B) $2\sqrt{10}$
- (C) $8\sqrt{2}$
- (D) $9\sqrt{3}$
- (E) $6\sqrt{5}$

Q3. $\sqrt{50} \cdot \sqrt{2} =$

- (A) 10
- (B) $5\sqrt{2}$
- (C) $10\sqrt{2}$
- (D) $2\sqrt{25}$
- (E) 25

Q4. A rainstorm causes the water level in a river to rise by $\sqrt{32}$ meters. Simplify this expression

- (A) $4\sqrt{2}$
- (B) $2\sqrt{8}$
- (C) $8\sqrt{2}$
- (D) $6\sqrt{3}$
- (E) $3\sqrt{5}$

Q5. $\frac{\sqrt{90}}{\sqrt{10}} =$

- (A) 3
- (B) $\sqrt{3}$
- (C) $\sqrt{9}$
- (D) $3\sqrt{2}$
- (E) $\sqrt{10}$

Q6. $\sqrt{75} - \sqrt{48} =$

- (A) $5\sqrt{3} - 4\sqrt{3}$
- (B) $3\sqrt{5} - 2\sqrt{6}$
- (C) $4\sqrt{2} - 3\sqrt{2}$
- (D) $5\sqrt{2} - 2\sqrt{5}$
- (E) $2\sqrt{6} - 3\sqrt{3}$

Q7. $\sqrt{20} \cdot \sqrt{5} =$

- (A) $10\sqrt{2}$
- (B) $2\sqrt{10}$
- (C) 10
- (D) $20\sqrt{2}$
- (E) 25

Q8. The area of an ecosystem is $\sqrt{98}$ square kilometers. Simplify this expression

- (A) $7\sqrt{2}$
- (B) $14\sqrt{2}$
- (C) $7\sqrt{5}$
- (D) $2\sqrt{49}$
- (E) $7\sqrt{14}$

Open Ended Questions (FRQ)

Q9. Simplify $\sqrt{108} + \sqrt{32} - \sqrt{50}$ and show all steps

Q10. A weather forecast predicts a temperature change of $\sqrt{128}$ degrees. Simplify this expression and explain your reasoning

Chef's Tips

- Read each question carefully
- Use the correct operations for radicals
- Check your work for accuracy